

Duy Tran

Machine Learning & AI Engineer | GStar Bootcamp 2026 Applicant

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PROFILE

M.S. Computer Science candidate at Georgia Tech with a 4.0 GPA and a machine learning specialization. Has built evaluated AI and ML projects across information retrieval, deep learning, and reinforcement learning using Python and PyTorch. Published a full-length AI-literacy book and built a Vietnamese semantic-search prototype with held-out ranking, latency, and robustness evaluation. Seeking to deepen from applied LLM projects into direct Transformer training, post-training, inference optimization, and agent evaluation.

EDUCATION & TECHNICAL TRAINING

Georgia Institute of Technology

Aug 2025 – Spring 2028 (Expected)

Master of Science in Computer Science, Machine Learning specialization | GPA: 4.0 / 4.0

Relevant coursework: Machine Learning, Graduate Introduction to Operating Systems

VietAI Foundation of Machine Learning

Dec 2020 – Apr 2021

10-week machine learning and deep learning program | Ranked top 3 in the class

Trained a CNN on approximately 10,000 images and an RNN for sentiment classification on approximately 3,000 Vietnamese restaurant reviews.

Earlier Education

University of Economics Ho Chi Minh City, B.B.A. | De Anza College, Liberal Arts & Sciences with Mathematics & Engineering emphasis

SELECTED AI & MACHINE LEARNING WORK

Tasco Semantic Search & Ranking (ts-semsearch)

github.com/duyhub/ts-semsearch | 2026

Python | FastAPI | PyTorch ecosystem | AWS Bedrock | Docker

- Built an intent-aware retrieval and ranking engine for Vietnamese point-of-interest search: normalized noisy queries, parsed intent, fused BM25 and bge-m3 dense retrieval with Reciprocal Rank Fusion, and re-ranked results using 9 interpretable signals.
- Achieved held-out NDCG@5 of 0.962 and Recall@3 of 0.983; hybrid retrieval beat BM25-only and dense-only baselines; 138/138 evaluation and adversarial robustness cases passed.
- Integrated Claude via AWS Bedrock for correction of degraded queries, with OpenAI fallback and Langfuse tracing, plus deterministic edit-distance typo correction.
- Packaged the FastAPI prototype in Docker and hosted a Railway project demo with local, local-first, and cloud modes, graceful degradation to a BM25-only floor, a published OpenAPI spec, and warm p95 latency of 9.3 ms.

Machine Learning Portfolio (CS7641)

Georgia Tech | 2026

Python | PyTorch | scikit-learn | cuML / CUDA

- Conducted four hypothesis-driven studies using a shared leakage-safe pipeline with stratified outer splits, train-only fitting and selection, cross-validation, fixed seeds, and held-out tests across imbalanced census and wine-quality datasets.
- Benchmarked Decision Tree, kNN, linear/RBF SVM, and MLP models, implementing MLPs in both scikit-learn and PyTorch; analyzed bias-variance behavior, learning curves, complexity curves, F1, PR-AUC, and Macro-F1.
- Optimized neural-network weights with randomized hill climbing, simulated annealing, and genetic algorithms; ran 7-way optimizer and 5-family regularization ablations under fixed compute budgets, with Nesterov reaching 0.6805 held-out Adult F1.
- Hand-implemented NumPy K-Means and GMM/EM, studied four dimensionality-reduction methods, and implemented DQN and Double DQN in PyTorch; tabular Q-learning matched DQN quality at 27–32× lower wall-clock on the studied tasks.

Hiếu AI để dùng cho đúng (Understand AI to Use It Right)

learningai.vn | 2026

Published book explaining how language models work, written for non-technical readers

- Authored and published a seven-part, 20-chapter book (roughly 74,000 words) explaining the mechanics of modern language models without programming or mathematics: tokens, next-token prediction, sampling and temperature, and the context window.
- Derived practice from those mechanics rather than from rules of thumb, showing why deliberate prompting, context management, tool use, and splitting work across agents behave the way they do, and why hallucination and sycophancy are predictable consequences of how the models are built.
- Turned those failure modes into judgment readers can apply: habits for verifying model output, guidance on where AI belongs in real work, and a way to reason about the claims made for the technology and the changes it brings.
- Traced every factual claim to a frontier-lab publication or peer-reviewed paper, verified by opening the source, and explained the concepts through 16 custom diagrams and a glossary; free to read, with an English edition in progress.

SYSTEMS ENGINEERING FOUNDATION

Graduate Operating Systems Portfolio

Georgia Tech | 2025

C / C++14 | pthreads | TCP sockets | gRPC | Protocol Buffers | POSIX IPC

- Built a C++ distributed file system with streaming gRPC services and weakly consistent client caches, coordinating concurrent clients through server-issued write locks, deadline handling, and inotify-driven asynchronous change notification.
- Built a boss-worker C file server over IPv4/IPv6 TCP and a caching web proxy backed by a separate daemon using POSIX shared memory, message queues, semaphores, and signal handlers that guarantee clean resource teardown.

EXPERIENCE

nftjam.io

Aug 2021 – Mar 2022

Co-Founder & Backend / Blockchain Engineer

- Built a RESTful escrow API brokering secure middle-wallet marketplace transactions for buying, selling, and listing assets.
- Contributed to Cardano's first multi-asset smart contract, cutting network fees and user wait times, scaling efficiently with the number of assets, and doubling revenue at launch.
- Analyzed and visualized market and trading-volume data in Python to inform leadership decisions.

LEADERSHIP

Guild Master, Albion Online

2020

Hardcore full-loot PvP MMORPG, roughly 30,000 concurrent and 300,000 daily active players

- Led a 400-member Vietnamese guild of players from widely different backgrounds to a top-30 global ranking within three months, competing against established international guilds in a high-pressure environment.

TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript

Machine Learning: supervised and unsupervised learning, reinforcement learning, deep learning (MLPs, CNNs, RNNs, DQN), information retrieval (BM25, embeddings, RRF, re-ranking), LLM integration, model selection and evaluation

ML Tooling: PyTorch, scikit-learn, cuML / CUDA, sentence-transformers, NumPy, Pandas, Matplotlib

Backend & Tools: FastAPI, REST, gRPC, Protocol Buffers, multithreading, TCP/IP, AWS Bedrock, Langfuse, Docker, Railway, Cloudflare Pages, Linux, Git